
Full Technical Report: Pedestrian Crossing Prediction using Deep Learning

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Github: <https://github.com/Kundu-Soham/DriveSense.git>

Google Colab: colab.research.google.com/drive/10teHBrmxWfdvdZtnmqPZtvkvryaMOh6?

Introduction

Autonomous vehicles require robust safety measures to navigate complex urban environments. A central challenge in autonomous navigation is predicting pedestrian crossing intent to allow timely, proactive vehicle responses. The goal of this project is to develop a deep learning model that predicts whether a pedestrian intends to cross the street using video sequences. This task is essential because accurate intent recognition prevents severe collisions while avoiding unnecessary traffic disruptions caused by overly conservative driving. Machine learning is uniquely suited for this problem because traditional rule-based models cannot easily process high-dimensional behavioral cues. Deep learning models automatically extract spatial visual patterns and temporal dynamics from video frames, enabling precise predictions from complex human behaviors.

Illustration

Figure 1 describes the overall architecture and workflow of the model in a simplistic manner.

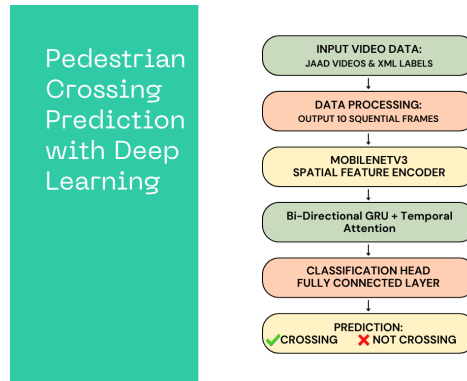


Figure 1: Overall architecture and workflow of the model.

Background & Related Work

Pedestrian intent prediction has been widely studied as a key component of autonomous driving systems. A comparative study using the JAAD dataset demonstrated the effectiveness of deep learning methods for predicting pedestrian crossing behaviour [1], while a recent vision-language foundation model approach showed that multimodal contextual information can further improve prediction performance [2]. Other work has proposed convolutional and recurrent architectures that jointly model pedestrian appearance, pose, and motion cues to estimate crossing intent from video sequences [3][4]. More recent approaches have incorporated temporal attention and transformer-based architectures to better capture long-term behavioural patterns and contextual information [5]. These studies provide a strong foundation for the project and motivate the use of deep learning models for pedestrian crossing intent prediction using the JAAD and PIE datasets.

Data Processing

Dataset sequences were extracted from raw Joint Attention in Autonomous Driving (JAAD) video clips and XML ground-truth annotations across a 60/20/20 video-level split into train, validation, and test sets as shown in Figure 2. Pedestrian tracks were verified, clipped to frame boundaries, cropped around target bounding boxes, and resized to 224×224 pixels. Variable track lengths were standardized to 10-frame sequences using linear index interpolation for longer tracks and oversampling for shorter tracks, with binary labels ($y \in \{0, 1\}$) assigned via majority-frame crossing intent. Figure 3 illustrates a sample processed sequence of frames from 1 video showing a pedestrian crossing.

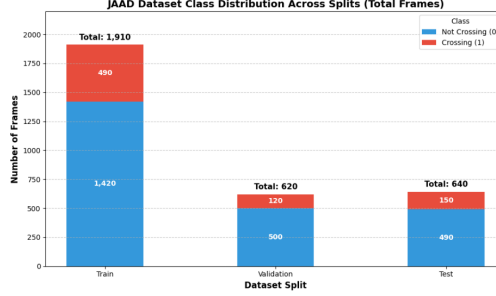


Figure 2: Overview and statistics of the JAAD dataset.



Figure 3: Sample visualization of the processed dataset.

Architecture

The proposed network combines spatial feature extraction, temporal sequential modeling, and a custom attention mechanism to predict pedestrian crossing intent. Sequential video frames are passed through a fine-tuned MobileNetV3-Small backbone to extract frame-level spatial embeddings. These feature vectors are processed by a 2-layer bidirectional GRU with a hidden dimension of 128 per direction, capturing complex temporal context across sequential observations. A custom temporal attention layer applies learnable scalar weights across time steps to dynamically emphasize critical frames prior to final representation aggregation. Finally, the weighted context vector is passed through a classification head comprising batch normalization, ReLU activation, 50% dropout, and a linear output layer to produce binary crossing probability logits.

Baseline Model

To serve as a comparative benchmark for the temporal network, a lightweight 2D Convolutional Neural Network baseline was constructed to perform single-frame spatial classification. The model architecture consists of two convolutional layers with batch normalization, ReLU activations, and max pooling, followed by a fully connected classification head with 50% dropout. Unlike the recurrent temporal pipeline, the baseline receives only the final image frame from each video sequence, resized to 224×224 pixels. The network is optimized using binary cross-entropy loss with the Adam optimizer at an initial learning rate of 0.01 over 20 epochs.

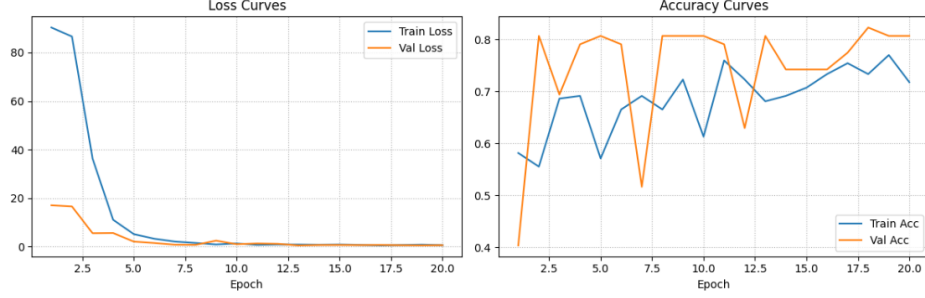


Figure 4: Training Curves of the Baseline Model.

Quantitative Results

To comprehensively evaluate model performance, training dynamics were tracked alongside fine-grained classification metrics on a 64 sequence test set. The training curves (Figure 5) reflect steady convergence over 31 epochs, with validation loss stabilizing near 0.5057 and validation accuracy peaking at 88.71%. On the test set, the regularized model selected from epoch 24, which provided the highest validation accuracy, achieved a test accuracy of 82.81%, a test loss of 0.8273. As detailed in Table 1, the class-specific breakdown shows robust performance on the majority non-crossing class with an F1-score of 0.89, while the minority crossing class achieved a precision of 0.643, recall of 0.600, and an F1-score of 0.621. Compared to the 2D CNN baseline which exhibited unstable training dynamics, the main sequence-based model achieved significantly smoother loss convergence and far superior decision stability by incorporating multi-frame temporal context.

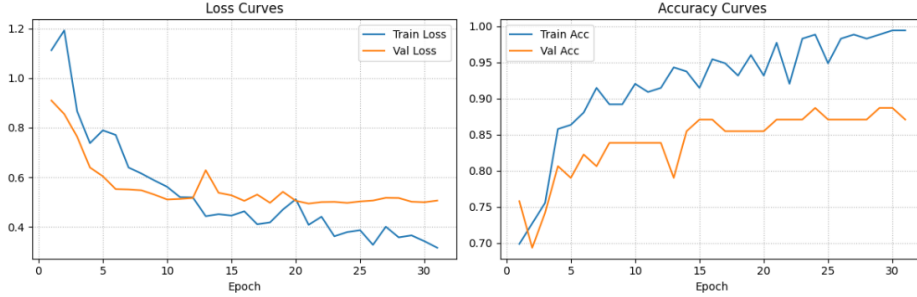


Figure 5: Training Curves of the Model.

Table 1: Test Results of the Trained and Optimized Model.

Class	Precision	Recall	F1-Score	Support
Non-Crossing	0.88	0.90	0.89	49
Crossing	0.64	0.60	0.62	15
Accuracy			83%	64
Macro Avg	0.76	0.75	0.75	64
Weighted Avg	0.82	0.83	0.83	64

Qualitative Results

To contextualize the quantitative findings, Figure 6 presents representative prediction sequences that were selected to highlight both successful classifications and the primary failure mode observed during evaluation. As shown in the top two rows, the model leverages multi-frame temporal attention to correctly identify non-crossing pedestrians, even when subjects walk near spatial clutter or exhibit pose variability. Conversely, the bottom row reveals a primary failure mode where a pedestrian

facing the curb triggers a false positive crossing prediction with 0.85 confidence due to forward posture, directional alignment, and limited observable motion. This tendency to predict crossing during curb-standing ambiguity directly explains why precision dropped to 0.643 on the minority crossing class in the quantitative results.

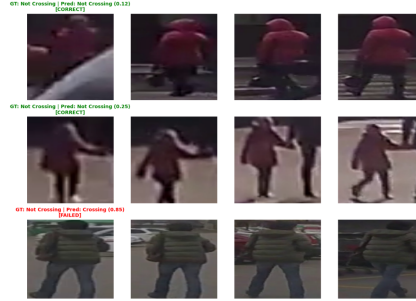


Figure 6: Results of Model's Output/Inference for Correct and Wrong Predictions.

Evaluating Model on New Data

To rigorously assess real-world generalization, the epoch 24 model checkpoint was evaluated on 20 completely unseen, held-out video sequences that were excluded from training, validation, and hyperparameter tuning. The network achieved 80.00% accuracy, an evaluation loss of 0.4587, and a strong ROC-AUC of 0.8533. The model maintained reliable prediction on non-crossing pedestrians with an F1-score of 0.88 across 15 sequences, but crossing recall dropped to 0.400 (2 True Positives and 3 False Negatives out of 5 crossing sequences). Although performance remained consistent with the primary test results, the reduction in crossing recall highlights the model's sensitivity to limited positive training samples and the increased variability present in previously unseen pedestrian behaviours.

Discussion

Overall, the spatio-temporal architecture performed strongly, significantly outperforming the single-frame 2D CNN baseline by using temporal attention to capture motion cues across sequential frames. The smoother training convergence and improved classification metrics indicate that incorporating temporal information enables the network to learn more discriminative pedestrian behaviour than spatial features alone. The primary limitation involved edge-case ambiguity where pedestrians standing stationary near the curb while facing forward triggered false positive crossing predictions. This visual ambiguity directly reduced precision and recall for the minority crossing class across both evaluation splits, suggesting that class imbalance and subtle behavioural differences remain challenging for the model. These findings demonstrate that while MobileNetV3-Small embeddings and bidirectional GRU networks learn effective dynamic representations, incorporating complementary cues such as pedestrian velocity, pose estimation, or additional contextual scene information would likely improve robustness and reduce ambiguous predictions in real-world autonomous driving systems.

Ethical Considerations

The JAAD and PIE datasets were collected in specific locations and traffic conditions, which may not fully represent pedestrian behavior across all environments, potentially introducing bias into the model's predictions. Additionally, incorrect intent predictions in autonomous driving systems could lead to unsafe decisions around pedestrians, highlighting the importance of thorough testing and ensuring that such models are used as part of a larger safety framework rather than as a standalone decision-making system.

References

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